

BEACONHOUSE NATIONAL UNIVERSITY



LAB 6

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Total Marks: /10

Due Date: 17 Mar, 2025

Lab Manual

For

COAL(SEC_B)

Course Instructor	Ms. Nimra Abbas
Lab Instructor	Ayesha Seher
Semester	Spring 2025

Problem No.1:

Explain the MOV instruction use cases using EMU8086?

Solution:

.MODEL SMALL

.STACK 100H

.DATA

.CODE

MAIN PROC

MOV AX, 10 ; AX = 10

MOV BX, 20 ; BX = 20

MOV AX, BX ; AX = 20

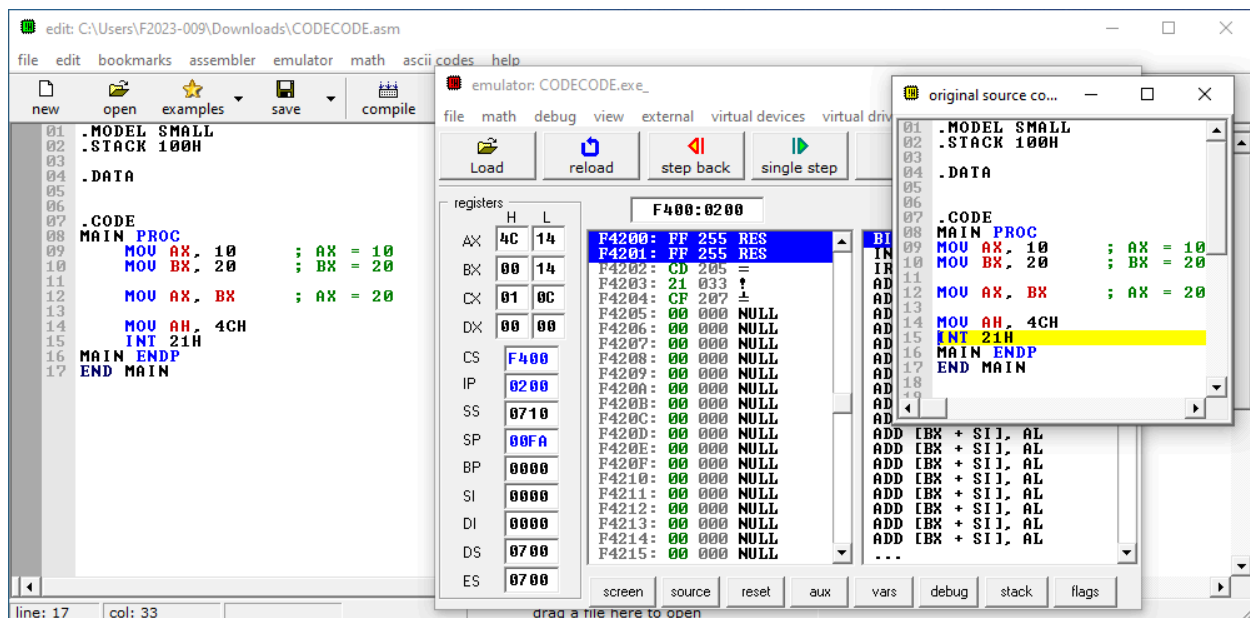
MOV AH, 4CH

INT 21H

MAIN ENDP

END MAIN

Screenshot:



Problem No.2:

Find the result of executing ADD AX, BX if AX = 0x1234 and BX = 0x1111?
0x2345 (0x1234 + 0x1111 = 0x2345).

Solution:

```
.MODEL SMALL
```

```
.STACK 100H
```

```
.DATA
```

```
    AXX DW 0x1234
```

```
    BXX DW 0x1111
```

```
.CODE
```

```
MAIN PROC
```

```
    MOV AX, AXX
```

```
    MOV BX, BXX
```

```
    ADD AX, BX ; AX = AX + BX -> 0x1234 + 0x1111 = 0x2345
```

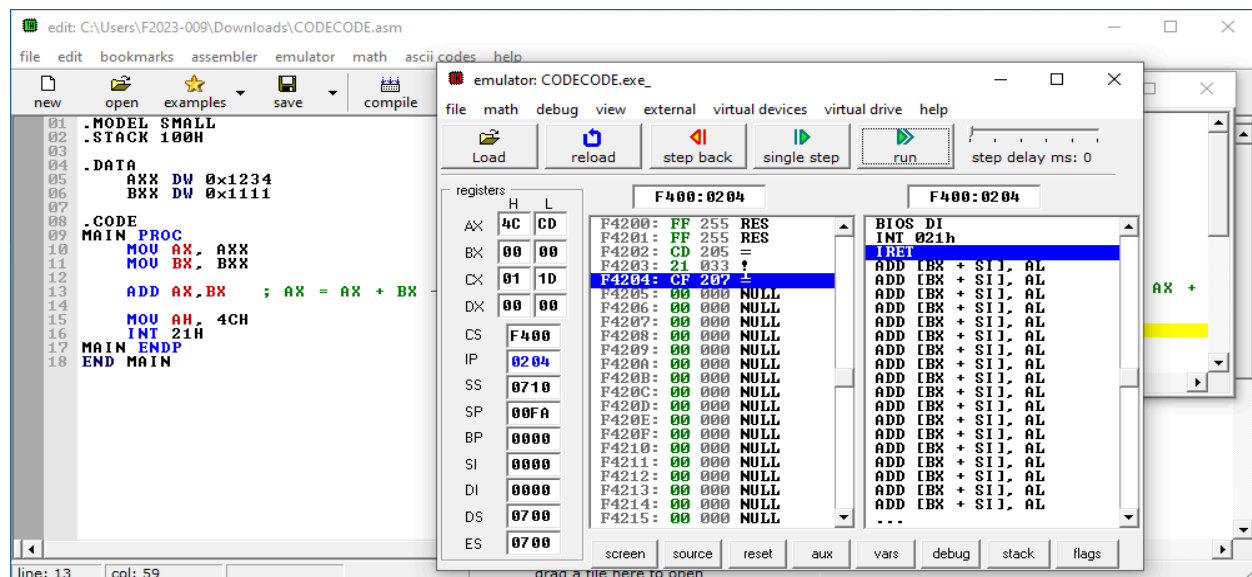
```
    MOV AH, 4CH
```

```
    INT 21H
```

```
MAIN ENDP
```

```
END MAIN
```

Screenshot:



Problem No.3:

Calculate the result of executing ADD AX, 0xFF00 if AX = 0x1100.

Solution:

.MODEL SMALL

.STACK 100H

.DATA

num DW 0x1100

.CODE

MAIN PROC

MOV AX, num

ADD AX, 0xFF00

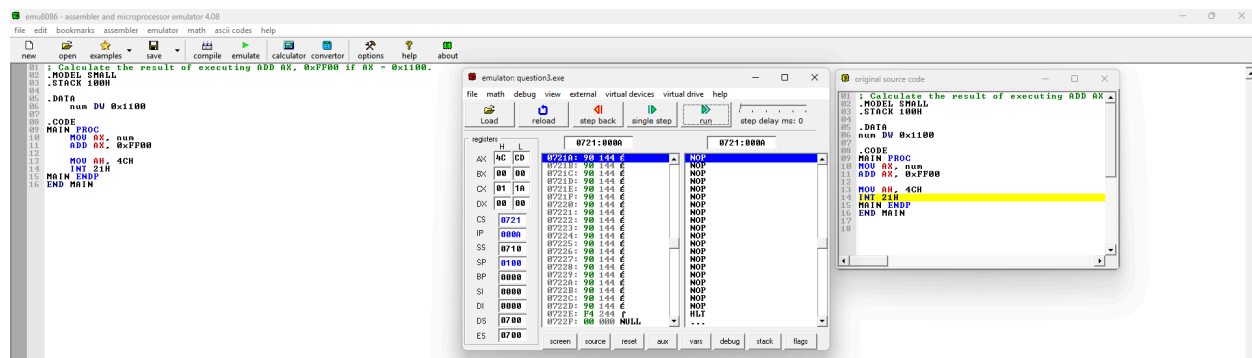
MOV AH, 4CH

INT 21H

MAIN ENDP

END MAIN

Screenshot:



Problem No.4:

Calculate the result of MUL CX if AX = 10 and CX = 2.

Solution:

.MODEL SMALL

.STACK 100H

.DATA

num1 dw 10

num2 dw 2

.CODE

MAIN PROC

MOV AX, num1

MOV CX, num2

MUL CX

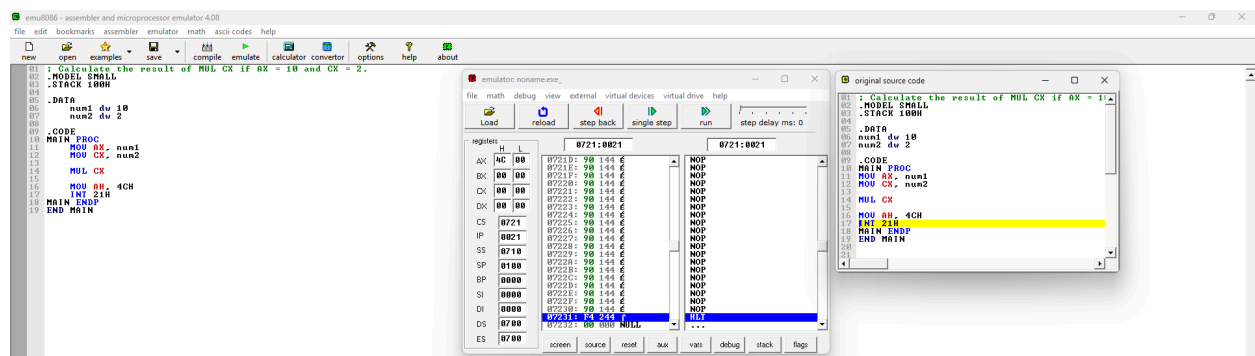
MOV AH, 4CH

INT 21H

MAIN ENDP

END MAIN

Screenshot:



Problem No.5:

Write a program to display multiples of 3.

Solution:

.MODEL SMALL

.STACK 100H

.DATA

message DB "Multiples of 3:", 0Dh, 0Ah, '\$'

multiples DB "3 ", "6 ", "9 ", "12", "15", "18", "21", "24", "27", "30", "\$"

.CODE

MAIN PROC

MOV AX, @DATA

MOV DS, AX

MOV DX, OFFSET message

MOV AH, 9

INT 21H

MOV DX, OFFSET multiples

MOV AH, 9

INT 21H

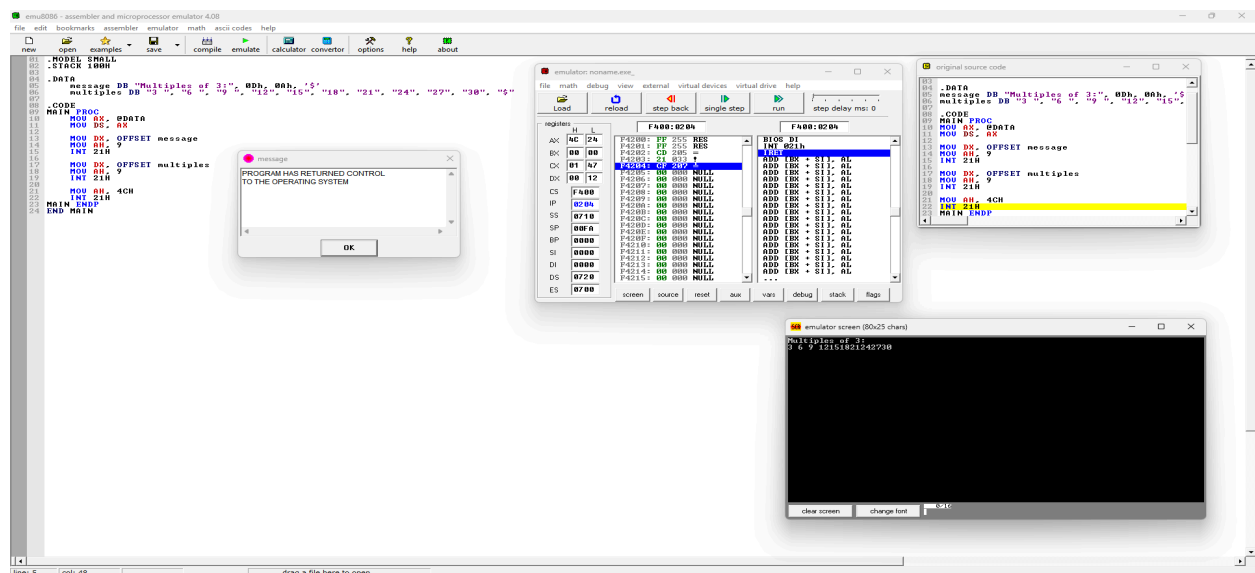
MOV AH, 4CH

INT 21H

MAIN ENDP

END MAIN

Screenshot:



Problem No.6:

If IMUL DX is executed and DX = -3, AX = 4, what will be the result.

Solution:

.MODEL SMALL

.STACK 100H

.DATA

num1 DW -3

num2 DW 4

.CODE

MAIN PROC

MOV AX, @DATA

MOV DS, AX

MOV AX, num2

MOV DX, 0

IMUL num1

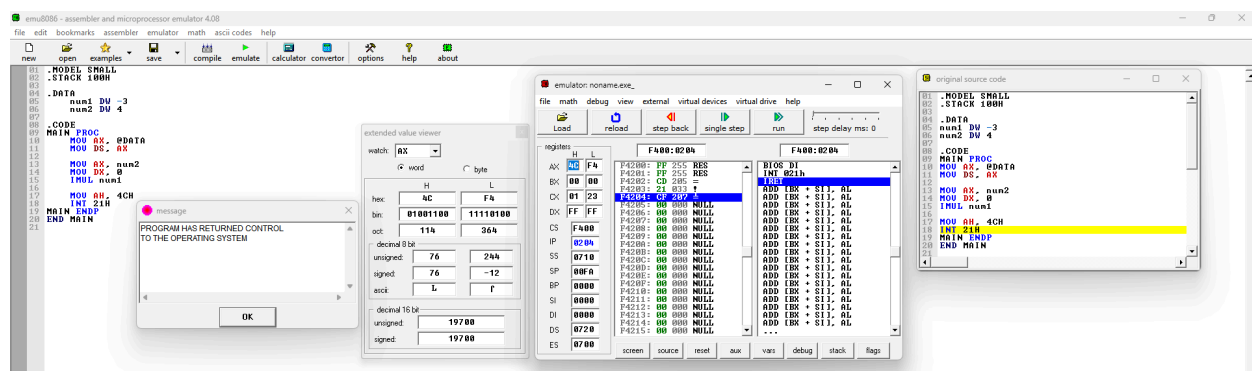
MOV AH, 4CH

INT 21H

MAIN ENDP

END MAIN

Screenshot:



Problem No.7:

Find the result of INC BX if BX = 0xFFFF.

Solution:

.MODEL SMALL

.STACK 100H

.CODE

MAIN PROC

MOV BX, 0FFFFH

INC BX ; BX = BX + 1 -> BX should become 0x0000

MOV AH, 4CH

INT 21H

MAIN ENDP

END MAIN

Screenshot:

